Exam

Databases, 24 May 2016

- Maximum number of points: 40. Score required to pass the exam (5.5): 22 points.
- For the true/false questions (questions 3 and 9) a simple true/false answer is sufficient.
- In general: keep your answers brief and to the point.

Exam questions

1. General

   What are the three levels of the ANSI-SPARC model? (2 points)

2. Conceptual Data Modelling

   Art sections of a museum are uniquely named after the type of art on display. A section has one or more curators. Each curator has a specialism. Curators are affiliated with at least one art section and are employees of the museum. Each employee has a name and a unique email address. Each art section has a section head (an employee). Art works have a collection ID, an artist, a title, and a year. Art works may be on display in an art section.

   (a) Draw an E/R model or a UML model for the case description above. Where needed, explain modelling decisions. (5 points)

   Tip: you do not need to model the concept of the museum as a whole.

   (b) Convert the model into a relational database schema. Where needed, explain design choices. (2 points)

   (c) General question: explain in one sentence the notion of 'role" in E/R relationships and UML associations. (1 point)

3. Relational model

   Indicate for each of the following statements whether it is true or false (0.5 point per statement):

   (a) The order of attributes in a relation schema is meaningful.

   (b) A superkey is the optimal candidate key.

   (c) A foreign key must be part of the key of the relation it is specified in.

   (d) The statement "The value of attribute `age` must be a positive integer" is an example of a database constraint.

4. Relational algebra

   Consider the database tables in Fig. 1.

   (a) Write a relational algebra expression for the query: "find the collection IDs of paintings that are being loaned to an exhibition in Rome". (2 points)

   (b) Assume you do a cross product of the relations `Paintings` and `Loans`. How many tuples and how many attributes would the resulting relation have? (2 points)

5. SQL queries

   Consider the database tables in Fig. 1.

   (a) Find paintings that have not been on loan, using the SQL construct `NOT IN`. (2 points)

   (b) Find the places (printed as "Locations") where paintings of Breitner have been on loan for an exhibition. (2 points)

   (c) Print (in alphabetical order) per artist the average insurance sum of her/his paintings. Include only artists of whom paintings have been loaned out. (2 points)

   (d) Explain what the query below will produce. (2 points)

```
SELECT P.title
FROM   Paintings P
WHERE  2 = ( SELECT COUNT(*)
             FROM   Loans L
             WHERE  P.colID = L.colID )
```

(e) Explain what the query below will produce. How many rows will the result have? (2 points)

```
SELECT P.colID, P.title, L.insurance
FROM   Paintings P LEFT JOIN Loans L
  ON   P.colID = L.colID AND L.insurance > 10000000
```

6. Consider a relation about wine tasting with the following pieces of information: taster (person who is tasting), tasted wine, region (of the tasted wine), taste notes (made by taster about a wine), favourite wine (of the taster). Which functional dependencies would you expect to hold? (2 points)

7. Consider the relation $R(A, B, C, D, E)$ with FDs $AB \to D$, $B \to C$, $A \to E$, and $E \to B$.

   (a) What is/are the minimal key(s) of this relation? (1 point)

   (b) Determine whether the relation is in Boyce-Codd Normal Form (BCNF) and, if not, what the violations are. (1 point)

   (c) Determine whether the relation is in Third Normal Form (3NF) and, if not, what the violations are. (1 point)

8. Consider the relation $R(A, B, C, D, E, F)$ with FDs $CD \to A$, $C \to B$, $D \to E$, and $B \to F$. The minimal key is CD.

   Use the decomposition method to split R into a set of subrelations that are in Boyce-Codd Normal Form. Use the first violation you encounter in the list of FD'S when decomposing. (3 points)

9. Indicate whether the following statements about database transactions are true or false (0.5 point per statement):

   (a) A "dirty read" is defined as the situation where one transaction reads an intermediate result of another transaction and the latter is subsequently aborted before committing.

   (b) 2 Phase Locking ensures that the schedule is conflict-serializable.

   (c) In multi-granularity locking the fact that one transaction has an IS (shared intension) lock on a particular table implies that another transaction cannot take an IX (exclusive intension) lock on that table.

(d) Multi-version concurrency control ensures that read-only transactions are never blocked.

10. Consider the schedule below, involving three transactions:

T1		R(C)		R(A)			W(E)	
T2	W(D)					R(A)		
T3			W(C)		R(B)			R(E)

Explain why the schedule is / is not conflict-serializable. In case it is serializable, list potential transaction orderings. (2 point)

11. Consider the schedule below, involving four transactions:

T1		W(A)						
T2			R(D)			R(B)		R(C)
T3				R(A)	W(B)			
T4	R(A)						W(C)	

(a) Assume T_1 aborts at the end. Are there other transactions that have to be rolled back? (1 point)

(b) Same question in case T_4 aborts. (1 point)

Paintings			
colID	artist	title	year
1	Metsu	Sick child	1664
2	Maes	Spinner	1652
3	Breitner	Singel bridge	1894
4	Breitner	Damrak	1903
5	Vermeer	Milk maid	1660
6	Steen	Prince's day	1670

Loans		
colID	exID	insurance
1	1	2,500,000
1	3	3,000,000
3	2	800,000
4	2	1,200,000
5	3	50,000,000
6	1	18,000,000
6	3	17,500,000

Exhibitions		
exID	year	place
1	2014	Berlin
2	2014	London
3	2015	Rome

```
Paintings(colID, artist, title, year)
Loans(colID -> Paintings, exID -> Exhibitions, insurance)
Exhibitions(exID, year, place)
```

Figure 1: Snapshot of a relational database of museum with information about paintings. The `Loans` table lists which paintings were loaned to which exhibitions (in places outside the museum) and for what price the painting was insured. NOTE: although this database is about a similar domain as the case description above, there is no formal connection.